Study Guide, Exercises, and Sample Questions

CSE 4615: Wireless Networks

Islamic University of Technology (IUT)

Lecture 6.1: IEEE 802.11 Wireless Local Area Networks

1. Topics Covered

- Scope of 802.11: the standard covers only OSI Layers 1–2, splitting the Data Link Control (DLC) layer into Logical Link Control (LLC) and Medium Access Control (MAC); 802.11 defines PHY transmission schemes and the MAC protocol, but no LLC functionality.
- Reference Model as an abstraction of hardware/software modules that lets developers discuss the standard; the overall 802.11 Architecture built from Basic Service Sets (BSS), Independent BSS (IBSS), Extended Service Sets (ESS), Access Points (AP), and the Distribution System (DS).
- Coordination Functions: the mandatory Distributed Coordination Function (DCF) used by all stations in a BSS, versus the optional Point Coordination Function (PCF) extension for QoS; IBSS as the simplest 802.11 network (minimum 2 stations, no station has priority, coordination distributed).
- Services: Station Services (SS) — MSDU delivery, (de)authentication, privacy — versus Distribution System Services (DSS) — (re)association, (dis)association, integration via the portal; DSSs are unavailable in an IBSS.
- The IEEE 802.11 WLAN family: 802.11b (2.4–2.5 GHz, up to 11 Mbps), 802.11a (5–6 GHz, up to 54 Mbps, OFDM), 802.11g (2.4–2.5 GHz, up to 54 Mbps, OFDM), 802.11n (dual-band, up to 600 Mbps, MIMO/frame aggregation/channel bonding), 802.11ac (5–6 GHz, up to 3.4 Gbps, MU-MIMO); all variants use CSMA/CA.
- 802.11 Frame Formats: the PLCP and PMD sublayers of the PHY; PLCP maps MPDUs into PSDUs; the frame finally transmitted over the channel is the OFDM PLCP frame; three MPDU types — management, data, and control frames — with variable frame-body size for data/management frames.
- Physical Layer technologies: the original three PHYs — 2.4 GHz Frequency Hopping Spread Spectrum (FHSS), Direct Sequence Spread Spectrum (DSSS), and Infrared (IR) — plus four later supplement PHYs; High Rate DSSS (HR/DSSS) using Complementary Code Keying (CCK) for higher data rates; OFDM as the dominant modern scheme.
- Listen-Before-Talk: Clear Channel Assessment (CCA) as physical carrier sensing using a power threshold (-82 dBm); the Network Allocation Vector (NAV) as virtual carrier sensing, a countdown timer that reserves the channel regardless of the medium's physical state.

- Distributed Coordination Function (DCF): CSMA-based sense-before-transmit with no collision detection (fading, hidden terminals make CD infeasible); Collision Avoidance (CSMA/CA) plus link-layer ACK/retransmission (ARQ) to cope with high bit-error rates.

- Interframe Spaces (IFS): SIFS, PIFS, DIFS, EIFS, and how aSlotTime and SIFS form the basis of all others; PIFS = SIFS + aSlotTime; DIFS = SIFS + 2×aSlotTime; shorter IFS gives higher medium-access priority; concrete 802.11a values (aSlotTime = $9\,\mu$s, SIFS = $16\,\mu$s, PIFS = $25\,\mu$s, DIFS = $34\,\mu$s, EIFS $\approx 200\,\mu$s).

- Collision Avoidance and the Backoff Procedure: random backoff after sensing the channel idle for DIFS; Contention Window (CW) mechanics, $CW = \min(2^{(n-1)}CW_{\min} - 1,\ CW_{\max})$, with $CW_{\min} = 16$, $CW_{\max} = 1024$ for 802.11a; binary exponential backoff on failed transmissions; frozen countdown resumption after a busy period (fairness for previously deferred stations).

- RTS/CTS exchange: reserving the channel with short Request-to-Send/Clear-to-Send frames to avoid colliding long data frames; the full DIFS + backoff + RTS + SIFS + CTS + SIFS + DATA + SIFS + ACK frame-exchange timeline; medium access overhead and DCF inefficiencies (idle channel time, collision probability, exponential CW growth under congestion).

- Recovery and Retransmission: Short Retry Counter (SRC) vs. Long Retry Counter (LRC) selection based on a size threshold (or RTS use); default limits SRC = 7, LRC = 4; frame discard on limit; counters reset on success.

- Post-backoff: a mandatory random backoff performed after every successful transmission (even with no queued MSDU), guaranteeing that every frame is delivered with backoff and reducing delivery delay under light load.

- Fragmentation: partitioning an MSDU into smaller MPDUs above a fragmentation threshold to reduce retransmission cost and raise success probability on error-prone channels, at the cost of overhead; defragmentation as reassembly; maximum MSDU length of 2346 bytes; only unicast-addressed MPDUs may be fragmented.

- Hidden Stations and RTS/CTS: how a station invisible to the sender but visible to the receiver is protected via NAV set from overheard CTS frames; SIFS-spaced RTS/CTS/Data/ACK sequences; why SIFS $<$ DIFS always privileges CTS and ACK.

- Synchronization and Cell Search: the Timing Synchronization Function (TSF) and common clock; the beacon frame and Target Beacon Transmission Time (TBTT); AP-only beacon generation in infrastructure BSS (delayed by PIFS if the channel is busy) versus distributed, contention-based beacon generation in IBSS (synchronizes to the fastest clock); beacon contents (BSSID, Beacon Interval, PHY parameters, CFP duration, available channels, power limits, traffic indication).

- Scanning Procedures: passive scanning (station listens for beacons up to MaxChannelTime) versus active scanning (station broadcasts Probe Request and waits for Probe Response); the Station Management Entity (SME) decides which mode to use; the general association sequence (scan $\rightarrow$ select AP $\rightarrow$ authenticate $\rightarrow$ associate $\rightarrow$ DHCP).

- Point Coordination Function (PCF): the Point Coordinator (PC, typically in the AP) grants priority access; PCF protected from DCF access by NAV; the superframe structure of a Contention-Free Period (CFP, PCF-controlled) followed by a Contention Period (CP, DCF-controlled); CF-Poll/CF-Ack piggybacking, CFP delay when the channel is busy at TBTT, and the CF-End frame signaling CFP termination.

- QoS Support Mechanisms of 802.11e: the Hybrid Coordination Function (HCF) with two access mechanisms — Enhanced Distributed Channel Access (EDCA, contention-based, CP only) and HCF Controlled Channel Access (HCCA, polling-based, both CP and CFP); QBSS and Hybrid Coordinator (HC) concepts; DCF as the basis for both PCF and the HCF's contention-based access.

- Contention-based (EDCA) Medium Access: four Access Categories (AC_VO, AC_VI, AC_BE, AC_BK), each an independent backoff entity;
  AIFS[AC] = SIFS + AIFSN[AC]×aSlotTime (AIFSN $\geq$ 2); per-AC $CW_i[AC] = \min[2^i(CW_{\min}[AC]+1)-1,\ CW_{\max}[AC]]$; smaller $CW_{\min}/CW_{\max}$ raises priority but also collision risk; per-AC retry counters (QSRC, QLRC) and maximum MSDU lifetime; virtual collisions resolved internally by priority.

- TXOP concepts: EDCA-TXOP (contention-won interval, bounded by TXOPlimit) vs. HCCA-TXOP / polled TXOP (HC-granted, NAV-protected); the Controlled Access Phase (CAP) during which the HC controls the medium.

- Controlled Medium Access (HCCA): HC obtains TXOPs after sensing the medium idle for PIFS without backoff; AIFSN[AC] chosen so the earliest EDCA access is never faster than legacy DIFS; QoS CF-Poll transmitted after PIFS with no backoff; polled stations may send multiple frames separated by SIFS within TXOPlimit.

- IEEE 802.11e Superframe: beacon-initiated superframe with CFP (only polled transmissions) followed by CP (HC may still poll, unlike legacy PCF); Block Acknowledgment (multiple MSDUs delivered in one TXOP, acknowledged together, improving throughput efficiency); Direct Link Protocol (DLP) enabling direct station-to-station transfer instead of always relaying through the AP.

- 802.11 Unicast vs. Broadcast handling: unicast uses RTS/CTS, ACK/retransmission, exponential backoff, and auto-rate adaptation; broadcast uses none of these and transmits at a single (low) rate.

- Radio Spectrum Management: 802.11b/g's 2.4–2.485 GHz band split into 11 overlapping channels with only 3 non-overlapping channels available (classic 1/6/11 reuse); 802.11a's 5–6 GHz band offering 8 non-overlapping OFDM channels; AP-side channel selection and the resulting inter-AP interference risk.

2. Focus Areas

- Being able to redraw the 802.11 architecture (BSS, IBSS, ESS, DS, AP, portal) from memory and correctly place DCF/PCF and SS/DSS within it.

- Deriving PIFS and DIFS from SIFS and aSlotTime algebraically, and plugging in the 802.11a numbers (aSlotTime = 9 μs, SIFS = 16 μs) to reproduce PIFS = 25 μs and DIFS = 34 μs without looking them up.

- Explaining, precisely, why 802.11 cannot use collision detection (CD) the way wired Ethernet does, and how CSMA/CA plus ARQ substitutes for it.
- Working the Contention Window formula $CW = \min(2^{(n-1)}CW_{\min} - 1,\ CW_{\max})$ across multiple failed-attempt counts n, and knowing at which attempt CW saturates at $CW_{\max} = 1024$ for 802.11a.
- Reconstructing the full RTS/CTS/DATA/ACK timing diagram with every SIFS/DIFS/backoff gap labeled, and explaining why RTS/CTS is often disabled despite solving the hidden-terminal problem.
- Working through a two-station (or two-AP) backoff race numerically — given two random backoff counts, determining who transmits first and what the other station does afterward (freeze vs. resume).
- Explaining the hidden-station scenario (a station that hears CTS but not RTS) and why NAV, not physical carrier sense, is what protects the exchange in that case.
- Contrasting passive scanning (beacon-driven, 1–2 message exchange) and active scanning (probe-driven, 2–4 message exchange) step by step, including which entity (SME) decides between them.
- Distinguishing the legacy PCF superframe (CFP then CP, single Point Coordinator, polling only) from the 802.11e HCF superframe (HC can poll during *both* CFP and CP).
- Understanding why EDCA uses four independent backoff entities (one per AC) inside a single station, how a "virtual collision" between two ACs in the same station is resolved, and why AC_VO gets the smallest AIFSN/CW.
- Being able to compute AIFS[AC] from SIFS, AIFSN[AC], and aSlotTime, and $\text{CW}_i[\text{AC}]$ from $CW_{\min}[AC]$ and the backoff stage i.
- Explaining TXOP, EDCA-TXOP vs. HCCA-TXOP, and the Controlled Access Phase (CAP) in your own words, with an example of what happens when a TXOPlimit is reached mid-delivery.
- Reproducing the fragmentation/defragmentation trade-off (lower retransmission cost vs. higher overhead) and stating the 2346-byte maximum MSDU length from memory.
- Explaining why 2.4 GHz 802.11b/g only offers 3 non-overlapping channels despite having 11 numbered channels, while 802.11a's 5 GHz band offers 8 non-overlapping channels, and the practical AP-deployment consequence of each.

3. How to Study These Topics

1. Redraw the 802.11 reference architecture from memory: BSS, IBSS, ESS, AP, DS, and the portal to a non-802.11 LAN; label DCF and PCF as concepts layered on top of it.
2. Build a two-column table from memory: Station Services (MSDU delivery, authentication, privacy) vs. Distribution System Services ((re/dis)association, integration); note which column is unavailable in an IBSS and why.

3. Derive PIFS and DIFS algebraically from SIFS and aSlotTime, then substitute the 802.11a numbers and check against 25 μs and 34 μs.

4. Write one paragraph, in your own words, on why 802.11 cannot do collision detection and how CSMA/CA plus link-layer ACK/retransmission compensates.

5. Practice the Contention Window recurrence $CW = \min(2^{(n-1)}CW_{\min} - 1,\ CW_{\max})$ for $n = 1$ through 8 with $CW_{\min} = 16$, $CW_{\max} = 1024$, and find the exact attempt at which CW first saturates at $CW_{\max}$.

6. Redraw the full DIFS + random backoff + RTS + SIFS + CTS + SIFS + DATA + SIFS + ACK timeline, labeling every gap, and separately redraw the no-RTS/CTS version (DIFS + backoff + DATA + SIFS + ACK).

7. Work a numeric two-station backoff race (e.g., Station 1 draws backoff = 15, Station 2 draws backoff = 25): determine who wins, what value the loser's counter holds when it resumes, and why that value is not re-randomized.

8. Write a short paragraph explaining the hidden-station example from the lecture (a station that hears the CTS but not the RTS) and why the NAV — not physical carrier sensing — is what keeps it silent.

9. Prepare a one-page handwritten comparison: legacy PCF (single PC, one CFP + one CP per superframe, DCF-protected by NAV) vs. 802.11e HCF (HC polls during both CFP and CP, four EDCA backoff entities in CP).

10. Practice computing AIFS[AC] = SIFS + AIFSN[AC]×aSlotTime for a small set of practice AIFSN values you choose, and separately compute CW_i[AC] for two or three backoff stages i.

11. Summarize, in one sentence each, SRC vs. LRC, post-backoff, fragmentation/defragmentation, Block Acknowledgment, and Direct Link Protocol (DLP) — five mechanisms that all exist to reduce wasted airtime or wasted retransmissions.

12. Draw the 2.4 GHz channel-overlap diagram (11 channels, only channels 1/6/11 non-overlapping) next to the 5 GHz 802.11a channel plan (8 non-overlapping channels), and write one sentence on the deployment implication of each.

4. Exercises

4.1. Architecture, Services, and Frame Formats

1. Define BSS, IBSS, and ESS in one sentence each, and explain how an ESS is built out of multiple BSSs.

2. Explain the difference between DCF and PCF as coordination functions, and state which one is mandatory and which is optional.

3. List the two categories of 802.11 services (SS and DSS) and give two examples of each. Which category is unavailable in an IBSS, and why?

4. A non-802.11 LAN device needs to exchange data with a station inside an 802.11 ESS. Name the 802.11 entity that performs this integration function, and describe the path the frame takes (which sublayers/entities it passes through).

5. List the three MPDU (MAC frame) types defined by 802.11 and state, for each, whether its frame body is fixed or variable length.

6. Compare 802.11b, 802.11a, 802.11g, 802.11n, and 802.11ac on frequency band and maximum data rate. Which of these five standards can operate in *both* the 2.4 GHz and 5 GHz bands simultaneously?

4.2. Physical Layer and Listen-Before-Talk (with Numericals)

7. Name the three original 802.11 PHY types and briefly describe the transmission principle behind each (FHSS, DSSS, IR).

8. Explain how HR/DSSS achieves a higher data rate than plain DSSS despite operating in the same band, referencing Complementary Code Keying (CCK).

9. The lecture states the CCA power threshold is -82 dBm. If a station measures received signal power of -78 dBm on the channel, is the channel idle or busy according to CCA? What if the measured power is -85 dBm?

10. Explain the difference between physical carrier sensing (CCA) and virtual carrier sensing (NAV). Give one scenario where CCA alone would fail to prevent a collision but NAV would succeed.

11. A station receives an RTS frame whose Duration field specifies that the full RTS–CTS–DATA–ACK exchange will take 312 μs. If the station starts its NAV timer the instant it finishes receiving the RTS, and 90 μs have already elapsed by the time you check, how much longer must the NAV countdown continue before the station may attempt to sense the channel again?

4.3. Interframe Spaces and Timing (with Numericals)

12. List the four Interframe Spaces (SIFS, PIFS, DIFS, EIFS) in increasing order of duration, and state which one gives a station the *highest* priority for medium access.

13. Using aSlotTime = 9 μs and SIFS = 16 μs (802.11a values), compute PIFS = SIFS + aSlotTime and DIFS = SIFS $+2\times$aSlotTime. Confirm your answers equal 25 μs and 34 μs.

14. Suppose a hypothetical 802.11 variant has aSlotTime = 20 μs and SIFS = 10 μs. Compute PIFS and DIFS for this variant using the same formulas.

15. Two frames are queued: one under PCF (PIFS access) and one under DCF (DIFS access), and both stations sense the channel become idle at the same instant. Using the 802.11a values from Question 14, how many microseconds sooner can the PCF station begin transmitting than the DCF station?

16. Explain, in one paragraph, when EIFS (rather than DIFS) is used, and why using a longer IFS in that situation is the correct design choice (connect it to the hidden-terminal problem).

4.4. Backoff, Contention Window, and Collision Avoidance (with Numericals)

17. Write the Contention Window formula $CW = \min\left(2^{(n-1)}CW_{\min} - 1,\ CW_{\max}\right)$ and define every symbol, including what "n" counts.

18. Using $CW_{\min} = 16$ and $CW_{\max} = 1024$ (802.11a values), compute CW for the 1st through 5th transmission attempts ($n = 1, \ldots, 5$). Confirm the 1st attempt gives $CW = 15$ (i.e., 15 slots, matching the lecture's RTS/CTS timing figure).

19. Continuing Question 18, find the smallest attempt number n at which CW first reaches (saturates at) $CW_{\max} = 1024$.

20. Duration of backoff is given by (Random Backoff count r) $\times$ aSlotTime, where $r =$ Rand$(0 \sim CW)$. If a station on its 3rd attempt ($n = 3$) draws the maximum possible backoff count, compute the resulting backoff duration in μs using aSlotTime $= 9\,\mu$s.

21. Two stations, A and B, both sense the channel idle after DIFS and independently draw random backoff counts from $CW = 15$ (1st attempt). Station A draws $r_A = 15$ and Station B draws $r_B = 25$ is *not* possible on the 1st attempt (why not?) — instead suppose Station A draws $r_A = 4$ and Station B draws $r_B = 9$. Using aSlotTime $= 9\,\mu$s, compute how many microseconds after the DIFS boundary each station would transmit if nothing interrupts the countdown, and state which station transmits first.

22. Continuing Question 21, once Station A transmits, Station B's countdown freezes. If Station B had counted down for 4 slot times before freezing, what is its remaining backoff count when the channel becomes idle again (for DIFS) after A's exchange? Explain why B does *not* redraw a new random value.

23. A station's first two transmission attempts fail. Using $CW_{\min} = 16$, compute CW for the 3rd attempt, and compute the *average* (expected) backoff duration for that attempt in μs (average $r = CW/2$, aSlotTime $= 9\,\mu$s).

4.5. RTS/CTS, Hidden Stations, and Medium Access Overhead (with Numericals)

24. Draw (in words, step by step) the full RTS/CTS-enabled frame exchange: DIFS, backoff, RTS, SIFS, CTS, SIFS, DATA, SIFS, ACK. Which of these gaps use SIFS and which use DIFS, and why does that difference matter for priority?

25. Explain the hidden-station scenario from the lecture: Station 6 cannot detect the RTS from Station 2, but can hear the CTS from Station 1. Explain step by step why Station 6 still refrains from transmitting.

26. A 1000-byte data frame is sent at 54 Mbit/s (802.11a's top OFDM rate). Using $T = \dfrac{L \times 8}{R}$, compute the raw data transmission time in μs (ignore PLCP preamble/header for this estimate).

27. Using your answer to Question 25, and the 802.11a values SIFS $= 16\,\mu$s, DIFS $= 34\,\mu$s, and an average 1st-attempt backoff of $CW/2 \times$aSlotTime $= (15/2) \times 9\,\mu$s, estimate the *total* channel-occupancy time for one successful RTS/CTS-protected data exchange, assuming RTS $= 20\,\mu$s, CTS $= 14\,\mu$s, and ACK $= 14\,\mu$s of airtime. (Sum: DIFS + avg. backoff + RTS + SIFS + CTS + SIFS + DATA + SIFS + ACK.)

28. Repeat Question 26 *without* RTS/CTS (DIFS + avg. backoff + DATA + SIFS + ACK). What percentage of the RTS/CTS-enabled exchange time in Question 26 is pure protocol overhead (i.e., everything except the DATA transmission time), versus for the non-RTS/CTS version?

29. The lecture states that at 10 active nodes, the probability of at least two nodes choosing the same backoff slot exceeds 40%. Explain, using the Binary Exponential Backoff formula, why this collision probability falls as CW_{min} increases, and why $CW_{\text{max}} = 1024$ acts as a ceiling on this benefit.

4.6. Recovery, Post-Backoff, and Fragmentation (with Numericals)

30. Explain the difference between the Short Retry Counter (SRC) and the Long Retry Counter (LRC): which frames increment which counter, and what are their default discard limits?

31. A station's frame exchange fails 7 times in a row for a short frame (SRC-governed). What happens to the frame on the 7th failure, given the default SRC limit of 7?

32. Explain post-backoff in one paragraph: why is it performed even when the station has no more MSDUs queued, and what benefit does it provide to a lightly loaded system?

33. An MSDU of 3000 bytes exceeds the 2346-byte maximum MSDU length limit stated in the lecture. Explain what must happen to this MSDU before it can be transmitted, and name the process that reverses it at the receiver.

34. If a 2346-byte MSDU is fragmented into MPDUs of at most 1000 bytes of payload each, how many fragments are required? Briefly explain one benefit and one drawback of using a smaller fragmentation threshold in a high-error-rate channel.

4.7. Synchronization, Beacons, and Scanning (with Numericals)

35. Explain the role of the Timing Synchronization Function (TSF) and the Target Beacon Transmission Time (TBTT) in an infrastructure BSS.

36. In an infrastructure BSS, if the channel is busy at TBTT, when is the beacon transmitted instead? Using PIFS = $25\,\mu$s (802.11a), if TBTT occurs at $t = 0$ but the channel only becomes idle at $t = 40\,\mu$s, at what time is the beacon transmitted?

37. Explain how beacon generation differs in an IBSS compared to an infrastructure BSS, and why an IBSS ends up synchronized to its *fastest*-running clock.

38. List, in order, the message exchange for passive scanning and for active scanning. Which mode involves fewer message types, and which mode lets a station discover an AP faster on a lightly loaded channel?

39. A station performs active scanning across 5 channels, spending at most MaxChannelTime on each. If MaxChannelTime = 20 ms per channel, estimate the maximum total scanning time across all 5 channels, and explain one factor from the lecture that could make the actual time shorter.

4.8. PCF and 802.11e QoS (EDCA/HCCA) (with Numericals)

40. Explain the legacy PCF superframe structure (CFP then CP) and state which coordination function is used in each period.

41. In the PCF example from the lecture, Station 3 detects the beacon and sets its NAV for the whole CFP, while Station 4 is hidden from the PC and does not set its NAV. Explain the consequence: is Station 4 correctly part of this PC-coordinated BSS? Justify your answer.

42. List the four EDCA Access Categories in increasing priority order and state which application each is intended for.

43. Using AIFS[AC] = SIFS + AIFSN[AC]×aSlotTime with SIFS = 16 μs and aSlotTime = 9 μs (802.11a), compute AIFS for a practice parameter set AIFSN = $\{7, 3, 2, 2\}$ assigned to AC_BK, AC_BE, AC_VI, AC_VO respectively. Which AC gets the shortest AIFS, and does that match its intended priority?

44. Using $CW_i[AC] = \min\left[2^i(CW_{\min}[AC] + 1) - 1,\ CW_{\max}[AC]\right]$ with a practice value $CW_{\min}[\text{AC\_VO}] = 3$, compute CW_0, CW_1, and CW_2 for AC_VO (i.e., $i = 0, 1, 2$).

45. Explain what a "virtual collision" is in EDCA and how a station resolves it internally when two of its own Access Categories reach a zero backoff counter at the same instant.

46. Explain the difference between an EDCA-TXOP and an HCCA-TXOP (polled TXOP), including which one is protected by NAV and which is bounded by TXOPlimit.

47. A station is granted an HCCA-TXOP and needs to send three MSDUs. Explain, step by step, how it can transmit all three within one TXOP using SIFS spacing, and what happens if the third MSDU would push the exchange past TXOPlimit.

4.9. Block Acknowledgment, DLP, and Radio Spectrum (with Numericals)

48. Explain how Block Acknowledgment improves throughput efficiency compared to acknowledging every MSDU individually.

49. Explain the purpose of the Direct Link Protocol (DLP) in 802.11e, and why relaying every frame through the AP (as in legacy 802.11) wastes channel capacity.

50. List two differences between 802.11 Unicast transmission and 802.11 Broadcast transmission from the lecture's comparison table.

51. The 2.4–2.485 GHz band is divided into 11 channels, but only 3 (channels 1, 6, 11) are non-overlapping. If two neighboring APs must both avoid interference and only these 3 non-overlapping channels are available, how many APs can be deployed in the same area without any two adjacent APs sharing a channel, assuming a simple linear (non-cellular) arrangement?

52. 802.11a offers 8 non-overlapping channels in the 5–6 GHz band, versus only 3 for 802.11b/g at 2.4 GHz. If a campus network needs to assign a unique non-overlapping channel to each of 6 closely spaced APs to avoid co-channel interference, is this possible using 802.11b/g alone? Is it possible using 802.11a alone? Justify both answers.

5. Sample Questions

5.1. Short Questions

1. What two OSI layers does the 802.11 standard cover, and into which two sublayers is the DLC layer split?
2. Define BSS, IBSS, and ESS in one sentence each.
3. What is the difference between DCF and PCF?
4. What are the two categories of 802.11 services, and which is unavailable in an IBSS?
5. Name the three original 802.11 PHY types.
6. What is the CCA power threshold stated in the lecture?
7. What is the difference between physical carrier sensing (CCA) and virtual carrier sensing (NAV)?
8. List the four Interframe Spaces in increasing order of duration.
9. What are aSlotTime, SIFS, PIFS, and DIFS for 802.11a?
10. Write the Contention Window formula and define $CW_{\min}$ and $CW_{\max}$ for 802.11a.
11. What problem does RTS/CTS solve, and what is one reason it is often disabled?
12. What is the maximum MSDU length in 802.11, and why might a station fragment an MSDU?
13. What is the difference between passive and active scanning?
14. Name the four EDCA Access Categories, from lowest to highest priority.
15. How many non-overlapping channels does 802.11b/g offer in the 2.4 GHz band, and how many does 802.11a offer in the 5 GHz band?

5.2. Descriptive Questions

16. Describe the complete 802.11 architecture (BSS, IBSS, ESS, AP, DS, portal) and explain how DCF and PCF relate to it.
17. Explain the full DCF frame-exchange procedure with RTS/CTS enabled, from a station sensing the channel idle for DIFS through receiving the final ACK, naming every interframe gap involved.
18. Explain the Contention Window / backoff mechanism in detail: how CW grows on failed attempts, how the random backoff count is drawn, and how a station resumes a frozen countdown after the channel becomes busy and idle again.
19. Describe the hidden-station problem and explain, with reference to NAV, how RTS/CTS solves it even for a station that can hear only the CTS and not the RTS.
20. Describe the legacy PCF superframe (CFP/CP) and contrast it with the 802.11e HCF superframe, explaining what changes for the Hybrid Coordinator's polling behavior.

21. Describe the EDCA mechanism in full: the four Access Categories, how AIFS[AC] and $CW[AC]$ are computed, what a virtual collision is, and how TXOPlimit bounds an EDCA-TXOP.

5.3. Analysis and Design Questions

22. A campus WLAN deployment must support both best-effort data traffic and latency-sensitive VoIP calls on the same 802.11e network. Propose an Access Category assignment for VoIP versus bulk file transfer, and justify your AIFSN/CW choices using the trade-offs discussed in the lecture.

23. Compare an RTS/CTS-enabled deployment against an RTS/CTS-disabled deployment for a densely populated lecture hall with many hidden-terminal pairs. Discuss the throughput trade-off and recommend which configuration is more appropriate, and why.

24. A network administrator is deploying three APs along a narrow office corridor using 802.11b/g, and separately considers deploying the same three APs using 802.11a. For each technology, discuss whether all three APs can be assigned non-overlapping channels, and what would happen if they could not.

25. Design a channel-access priority scheme (using the 4 EDCA Access Categories) for a station that simultaneously needs to send: (a) a VoIP packet stream, (b) interactive video conferencing, (c) a web browsing session, and (d) a background file backup. Justify your AC assignment for each using the AIFSN/$CW_{\min}$/$CW_{\max}$ trade-offs from the lecture.

26. A station repeatedly experiences failed transmissions in a congested BSS with 10 active nodes. Using the Binary Exponential Backoff formula and the lecture's collision-probability discussion, explain what happens to this station's throughput as contention increases, and propose one protocol-level mitigation (e.g., RTS/CTS, smaller fragmentation threshold, or 802.11e QoS) with justification.

27. Discuss why the lecture presents PCF and 802.11e HCF as successive attempts to add QoS on top of the same underlying DCF. Identify at least three specific limitations of PCF that HCF/EDCA addresses, and explain the protocol-design consequence of each.